



# A thermo-electro-mechanical simulation model for hot wire cutting of EPS foam



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## ABSTRACT

A one-dimensional thermo-electro-mechanical mathematical model describing the effects taking place within a Ni–Cr20% wire used in a hot-wire cutting process for free forming and rapid prototyping of expanded polystyrene (EPS) is investigated and simulated. The model implements and solves three semi-coupled non-linear differential equations (the heat diffusion equation, the electrical diffusion equation and the static equilibrium equation) with temperature dependent parameters in order to predict the temperature, kerfwidth, longitudinal stress and displacement, and other process parameters during cutting of EPS in contact with a cutting tool made of an electrically heated metal wire attached to a robot device. The finite difference method is used to solve the coupled equations in the two environments (domains) in which the hot-wire operates, namely air and EPS. The model is calibrated against experimentally obtained data. Novel findings are a transient temperature-dependent kerfwidth prediction and a relation between kerfwidth and the cutting angle as measured from the horizontal direction. These are important relations in the aim for higher geometrical accuracy of the hot-wire cutting process.

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## 1. Introduction

Since the late 1980s, the rapid prototyping and manufacturing (RP & M) industry has dramatically increased in size and scope due to the development of the 3D computer aided design (CAD) and large scale CNC machines. The Rapid prototype objects are often applicable in many fields such as marketing, design and sculpturing, casting, metallurgy, and for this reason the drive/market for them is increasing. Water jet cutting, 3D printing, laser and plasma cutting, electrical discharge machining and wire sawing are only a few of the RP & M processes that are commercially used. In the present work, we are focusing on the hot-wire cutting RP & M process, which on its own can be divided in two main groups: Layer Object Manufacturing (LOM) and Free Form Sculpturing (FFS). Both of these approaches are used for cutting soft materials that melt at low temperatures such as: polystyrene, wax and rubber. However, due to low price and availability, the materials most commonly used in hot-wire cutting processes are expanded polystyrene foam (EPS) and extruded polystyrene (XPS).

The first publications in the field of hot-wire cutting can be traced back to the works of [1–4], where simplified systems for cutting and sculpturing different materials including polystyrene were developed and tested. As a natural part of that development,

mathematical models for cutting of polystyrene foam using robotic CNC machines started emerging in the literature a little more than a decade ago [5–7], where the methodology is on modeling the process primarily by solving the 3 dimensional heat conduction equation with a heat source term responsible for the wire resistance heating. The most important parameter in the process is the linear heat input, that describes how much energy is transferred from the hot wire to the material per traveled unit length in the cutting direction. This parameter is moreover sometimes normalized with respect of the length of the wire. The research points out to a number of empirical correlations relating cutting parameters and effective heat input, which can be used in practice to describe the different effects/phenomena in the hot wire cutting of EPS/XPS during processes related to Layer Object Manufacturing and Free Form Sculpturing to be found in the works of [8–12]. The greatest findings of the latest group [10–12] are the volumetric melting energy of EPS/XPS during hot-wire cutting and an innovative acoustic feedback control. In all the work in the field, little effort was put into understanding the convective effects taking place in the contact between hot-wire and melting foam. [13] was the first example of investigating this effect by applying well known fluid correlations for natural convection in order to predict the heat transfer coefficient and by this approximate the degree of EPS melting.

Until now the focus in modeling of hot wire cutting has primarily been on the melting and forming of the EPS material, while the different wire dimensions and parameters (f.ex linear heat

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input) are taken into account by individual correlations. The objective of this paper is however to investigate the hot wire cutting process from a different perspective, namely by introducing a novel numerical model for hot-wire cutting describing primarily the cutting tool and its properties, during hot wire cutting of EPS. We believe that understanding the cutting tool behavior and its effect on the net material removal is a key element in understanding the process. The proposed thermo-electro-mechanical model aims to describe some of the multi-physical effects taking place during wire heating and foam cutting, and for this purpose the governing set of differential equations are presented and discretized henceforth.

## 2. Modeling and governing equations

The model consists of 3 differential equations, i.e. the one-dimensional heat diffusion equation used for finding the temperature in the wire, the one-dimensional electrical diffusion equation calculating the voltage distribution in the wire and the one-dimensional equilibrium equation including thermal strain for finding the displacement of the wire due to mechanical forces and thermal expansion. In mathematical terms the equations are described as:

$$\rho c_p(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left( k(T) \frac{\partial T}{\partial x} \right) + \dot{Q}_e''' + \dot{Q}_{loss}''' \quad [\text{W/m}^3] \quad (1)$$

$$\frac{\partial}{\partial x} \left( \sigma(T) \frac{\partial V}{\partial x} \right) = 0 \quad [\text{A/m}^3] \quad (2)$$

$$\frac{\partial}{\partial x} \left( E \left( \frac{\partial u}{\partial x} - \alpha \Delta T \right) \right) = 0 \quad [\text{Pa/m}] \quad (3)$$

where  $\rho$ ,  $T(t, x, V, \dot{Q}_e''', \dot{Q}_{loss}''')$ ,  $x$ ,  $t$ ,  $V(t, x, T)$ ,  $E$ ,  $u$ ,  $\Delta T$ ,  $\alpha$ ,  $\dot{Q}_e'''(t, x, V, T)$  and  $\dot{Q}_{loss}'''(t, x, V, T, \text{speed})$  are wire density, wire temperature, position in direction of the length of the wire, time, voltage, Young's modulus, mechanical displacements in  $x$  direction, temperature difference taking the initial configuration as reference, coefficient of linear thermal expansion, volumetric specific energy gained due to electric resistance heating and volumetric specific energy lost to the surroundings, respectively. The terms  $k(T)$ ,  $\sigma(T)$  and  $c_p(T)$  denote heat conductivity, electrical conductivity and specific heat capacity all being dependent on temperature. Temperature, voltage,  $\dot{Q}_e'''$  and  $\dot{Q}_{loss}'''$  are coupled between the equations. Note that dependency of speed is given by Eqs. (5)–(6) and hence it depends on whether we consider domain 1 or domain 2, see Fig. 1.

The energy gained in the wire due to resistance heating, i.e.  $\dot{Q}_e'''$  is expressed in terms of the wire conductivity, electric field intensity and the voltage drop through the wire as:

$$\dot{Q}_e''' = \sigma(T) E^2 = \sigma(T) \left( \frac{\partial V}{\partial x} \right)^2 \quad [\text{W/m}^3] \quad (4)$$

where  $E$  is the electric field intensity. In Eq. (4),  $\dot{Q}_e'''$  is proportional to the voltage drop squared, due to the definition of resistance heating, and thus a non-linearity is introduced in Eq. (1). On the other hand, the energy lost to the surroundings ( $\dot{Q}_{loss}'''$ ) depends mainly on the transverse boundary conditions at the different domains. At different places, in its transverse direction, the wire is in contact with two different environments, i.e. air and EPS, denoted – domain 1 and 2, respectively. The volumetric heat loss term in domain 1 is derived from the Newton law of cooling, while the loss of heat due to melting of EPS at the surface of the wire, is

derived in terms of wire motion, EPS energy parameters and kerfwidth-temperature relation, i.e.:

$$\dot{Q}_{loss}''' = h dA_c (T_{air} - T) / dV_m \quad \text{Domain 1} \quad [\text{W/m}^3] \quad (5)$$

$$\dot{Q}_{loss}''' = H_f \rho_f S F_{kerf}(T) dx / dV_m \quad \text{Domain 2} \quad [\text{W/m}^3] \quad (6)$$

where  $dA_c$  is the locally evaluated area of circumference,  $T_{air}$  temperature of air,  $dV_m$  locally evaluated wire volume,  $H_f$  specific melting energy of EPS foam,  $\rho_f$  density of EPS foam,  $S$  speed of motion of the wire during cutting,  $F_{kerf}(T)$  kerfwidth as a function of local temperature determined based on an empirical relationship obtained a priori to doing the simulations (see Section 4),  $dx$  locally evaluated wire length and  $h$  the total heat transfer coefficient described as:

$$h = h_{nc} + h_{rad} \quad (7)$$

$$h_{nc} = (Nu k_{af}) / D_w, \quad \text{where } T_f = 0.5(T + T_{air}) \quad (8)$$

$$h_{rad} = \epsilon \theta (T^2 + T_{air}^2) (T + T_{air}) \quad (9)$$

where  $h_{nc}$  is the convective heat transfer coefficient for air,  $h_{rad}$  the radiation heat transfer coefficient,  $Nu$  Nusselt number,  $k_{af}$  the thermal conductivity of air at the film temperature  $T_f$ ,  $\epsilon = \epsilon(x)$  the emissivity of the wire,  $\theta$  the Stefan-Boltzmann constant. The heat transfer coefficient  $h_{nc}$  is assumed to follow the Mikheyev correlation [14], as the choice is based on the work of [15]. The expression for  $\dot{Q}_{loss}'''$ , in both of the domains (inside the EPS – domain 2, and outside, i.e. in free air – domain 1), is evaluated locally, since temperature, speed and domain length may vary in  $x$  direction. The boundary conditions applied for the differential Eqs. (1)–(3) are as follows:

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 \quad \left. \frac{\partial T}{\partial x} \right|_{x=L} = 0 \quad (10)$$

$$V_{x=0} = V1 \quad V_{x=L} = 0 \quad (11)$$

$$u_{x=0} = 0 \quad \nu_{x=L} = 50 \text{ MPa} \quad (12)$$

where  $V1$  is a boundary condition describing an applied voltage of a particular size, while  $\nu$  denotes applied stress in direction  $x$ . The applied stress at the end of the wire is needed to keep the wire straight and by thus ensure that the desired cutting-results will be achieved.

## 3. Numerical method

In order to solve Eqs. (1)–(3), the finite difference method with a formulation similar to the one given in [16] is applied. The one dimensional space is discretized in  $N=120$  elements, each element of length  $\Delta x$  having a volume of  $\Delta V_m = A_s \Delta x$ , where  $A_s$  is the cross sectional area of the wire, while the time step is denoted  $\Delta t$ . The heat conduction equation is then discretized as:

$$H_i(T_i^{t+\Delta t} - T_i^t) = Hx_i(T_{i-1} - T_i) + Hx_{i+1}(T_{i+1} - T_i) + \dot{Q}_e'' + \dot{Q}_{loss}'' \quad [\text{W/m}^2] \quad (13)$$

$$Hx_i = \frac{1}{\frac{\Delta x_{i-1}}{2k_{i-1}(T)} + \frac{\Delta x_i}{2k_i(T)}} \quad H_i = \Delta x_i \rho c_p(T) / \Delta t \quad (14)$$

where the subscript, in the equations, is the space index in terms of  $i$ , and provides information about the space position, while superscript  $t$  and  $t + \Delta t$  denote current time and time after one time step, respectively. The temperature dependent properties are read from a table providing the appropriate value at a specific temperature. The heat loss to the surroundings is discretized as:

$$\dot{Q}_{loss} = h \Delta A_c (T_{air} - T_i) / A_s \quad \text{Domain 1 [W/m}^2] \quad (15)$$

$$\dot{Q}_{loss}'' = H_f \rho_f S F_{kerf}(T_i) \Delta x_i / A_s \quad \text{Domain 2 [W/m}^2] \quad (16)$$

where the discrete area of circumference is given as  $\Delta A_c = \pi D_w \Delta x$  and the remaining parameters are provided in Table 1. The boundary conditions are applied in 2 ghost points, one point at each side of the geometry.

Eqs. (2) and (3) are discretized in a similar manner, however the voltage and the displacement are evaluated at the boundaries of each element, due to the nature of physics applied. This is the well-known staggered grid arrangement which is widely used in CFD problems solved by the Finite Volume Method (FVM). The finite difference expression of Eq. (2) is written as:

$$LHx_i(V_{i-1} - V_i) + LHx_{i+1}(V_{i+1} - V_i) = 0 \quad [\text{A/m}^2] \quad (17)$$

$$LHx_i = \frac{1}{LR_{i-1} + LR_i} \quad (18)$$

$$LR_i = \frac{\Delta x_i}{2\sigma(T)} \quad (19)$$

Knowing the voltage distribution over the wire, the energy produced by resistance heating can be found as:

$$\dot{Q}_e'' = \sigma(T) A_s \frac{(\Delta V)^2}{\Delta x_i} \quad [\text{W/m}^2] \quad (20)$$

$$\Delta V = \frac{V_{i+1} - V_{i-1}}{2} \quad [\text{V}] \quad (21)$$

where  $\Delta V$  is discretized with a central difference. Due to the particular formulation of Eq. (13) and consequent change of units from  $\text{W/m}^3$  (in Eq. (1)) to  $\text{W/m}^2$  (Eq. (13)), Eq. (20) and (14) are also influenced, i.e. note the single  $\Delta x_i$  in the denominator of Eq. (20).

The finite difference formulation of Eq. (3) is found to be:

$$\frac{E_{i+1}}{\Delta x_{i+1}} u_{i+1} - \frac{(E_{i+1} + E_i)}{\Delta x_i} u_i + \frac{E_i}{\Delta x_{i-1}} u_{i-1} = E_{i+1} \alpha \Delta T_{i+1} - E_i \alpha \Delta T_i \quad (22)$$

Eq. (22) is a generalized formulation of the discretized equation (3), since the Young's modulus  $E_i$  has index  $i$  relating to every discretized element in the wire. However, in our problem the Young's modulus is constant through the entire geometry and  $E$  can be set as a constant. On the other hand,  $\Delta T_i$  denotes the temperature difference at element number  $i$ ,  $i \in [1: N]$

Eq. (13) is solved with an explicit finite difference scheme, while Eqs. (17) and (22) are solved with an implicit method. The coupling between the equations can be described by the following sequence:

1. Solve Eq. (13) based on old(or initial) values for temperature and voltage. Result – new temperature.

2. Solve Eq. (17) based on the new temperature found at the previous step. Result – voltage.
3. Solve Eq. (22), based on the new temperature found at Step 1. Result – displacement.
4. Repeat the 3 previous steps in the same order, while recently obtained values for temperature and voltage will now become old values and hence serve as initial conditions for Step 1.

This method of coupling the differential equations is sometimes referred to as semi-coupling, since we do not iterate between the different steps (1–3) in order to find a common solution, but rather use the solution of one equation to find the solution of the other equation at the next computational step. This method of solving the equation is appropriate and possible, only because the velocity of the current flow is fast (i.e. close to the light speed) and subsequently the voltage distribution in the wire in practice adjusts itself instantly and hence no iteration is needed.

Temperature dependent data for  $c_p(T)$  and  $\sigma(T)$  is taken from [17], while the temperature conductivity  $k(T)$  is assumed to follow the Smith–Palmer relationship as given in [18] and described by the following expression:

$$k(T) = 0.283 L_o T / \sigma(T) + 10.2 \quad (23)$$

where  $L_o$  is the Lorentz number and is approximately equal to  $2.45 \times 10^{-8} [\text{W}\Omega/\text{K}^2]$ .

#### 4. Experimental investigation

One of the biggest challenges in modeling of EPS cutting is describing the time dependent kerfwidth as a function of other cutting process parameters. The kerfwidth produced by the hot-wire cutting of the EPS material is a result of a complex heat transfer process combining conductive, convective and radiative heat transfer, involving combined melting and evaporation of EPS (which is 95% air). Due to the complexity of the problem, no modeling tool to investigate the process in depth has yet been given in literature and most of the work found has approached the problem experimentally. A number of authors have tried to express the kerfwidth as a function of different parameters, i.e: the applied current [11,9], the linear heat input [6,11,9,10] and the temperature [9]. Most of the authors have tried to work with the linear heat input, since it expresses how much heat per unit length in the cutting direction is transferred into the material and by this it determines to a large extend the kerfwidth during steady-state cutting. On the other hand, the linear heat input, by the literature's definition, is independent of (the wire) temperature and thus it is incapable of providing information about the transient-wire-cutting conditions,<sup>1</sup> nor whether the temperature of the wire is high enough for melting of the EPS to take place. To address these challenges, in the current work we express the kerfwidth as a function of the wire temperature ( $F_{kerf}(T)$ ), while at the same time we introduce a new parameter- the linear heat loss term (see Eq. (6)) that takes into account some of the cutting parameters as speed and melted zone dimensions (normally part of the linear heat input). In our formulation of the linear heat loss, we describe the affected melted zone in the EPS by means of the kerfwidth

<sup>1</sup> Transient-wire-cutting happens when the temperature of the wire is significantly changing during the cutting. This is normally observed at the start of the cutting, when the wire is brought into contact with the EPS material. Initially the wire is pre-heated in air and its temperature is relatively high, so at the start of the cutting much of the heat is transferred to the EPS in order to melt it and thus the wire is rapidly cooling at the first few centimeters of the cutting until a steady-state equilibrium cutting is reached. Transient-wire-cutting can also take place if some of the cutting parameters (current, voltage, speed) change during the cutting.

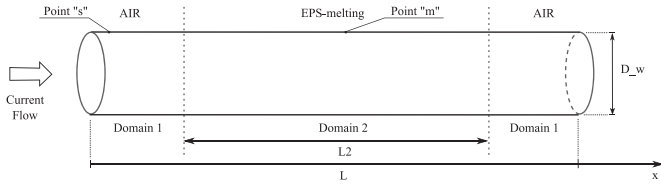


Fig. 1. Geometry of the wire applied in the one dimensional mathematical model.

function, which is dependent on the local wire temperature. This formulation of the EPS melted zone, we believe, is more accurate than the previously proposed one by [4], since it admits the kerfwidth to change during cutting operations and does not constrain the kerfwidth as being equal to the wire diameter, an assumption theoretically possible, but difficult to be kept in practical applications due to high friction forces.

A number of experiments were conducted for finding the relation between the kerfwidth and temperature for the particular material and tools in use. The experiments were performed using a robotic system with an integrated hot-wire tool for cutting polystyrene. A high-end IR camera FLIR SC7600 was attached to the tool for temperature measurements with an acquisition rate of 5 Hz during the cutting procedures. Pieces of EPS type 150 with dimensions  $1.5 \times 0.6 \times 0.1$  m were used during the experiments as samples for cutting. Special cylindrical holes ( $D_h = \varnothing 10$  mm,  $H_{depth} = 75$  mm) were cut in the EPS material for the purpose of visual exposure of the wire during cutting and subsequently performing IR temperature measurements. A sketch of the EPS samples with hole position and dimensions is shown in Figs. 2–4. Hot-wires with different diameters [0.5–0.9] mm were used in the experiments, as the wires were painted with a highly-durable black paint of known emissivity for accurate temperature reading. The cutting was performed according to the diagrams in Figs. 2–4, as each sample was cut twice and the middle piece was used for kerfwidth measurements. Figs. 5 and 6 show an example of the temperature measurements done during experiments, while Fig. 7 shows the middle piece cutting related to the kerfwidth measurement. A simple relation between kerfwidth and temperature was established as shown in Fig. 8.

Using the IR camera the temperature during cutting is measured at the holes of the EPS samples. Since the holes are small in size in comparison with the samples, a very small part of the camera recording is meaningful for the temperature determination, see Fig. 5.

In Figs. 5 and 6 the temperatures close to the beginning and the end of the time interval show the wire temperature in air, i.e. before the cutting of the sample is started and after it is finished, while the peaks in between present the temperature readings during cutting, when the wire is inside the cut holes. By removing the unimportant information in Fig. 5 and condensing the useful data, a better representation of the temperature reading is given in Fig. 6, where only the temperatures measured close by the holes

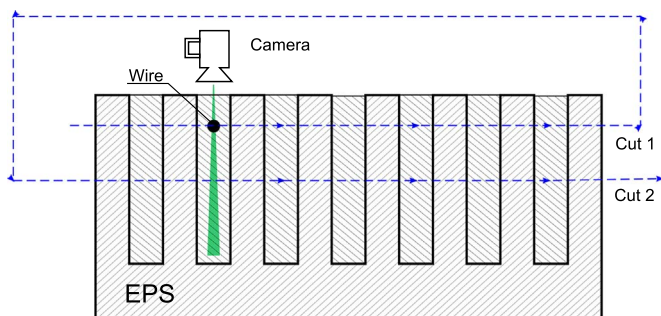


Fig. 2. Schematic diagram of the hot-wire cutting path for two sequential cuts.

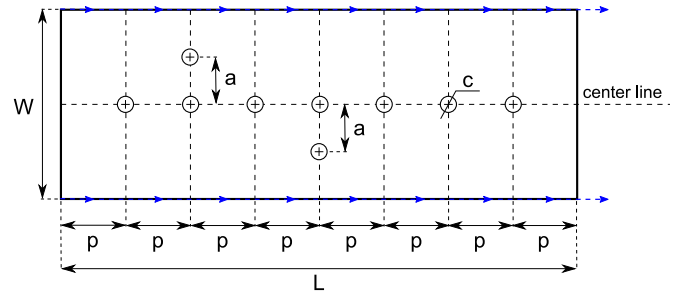


Fig. 3. Position of circular holes on the EPS block, as  $p=200$  mm,  $a=30$  mm,  $c=10$  mm. View from top of the EPS block. The hole dept is shown on Fig. 4. The lines with arrows indicate the direction of cutting. Dimensions  $W$  ( $=600$  mm) and  $L$  ( $=1.5$  m) refer to EPS block width and length.

are presented. The seven peaks (both graphs) show the temperature measurement inside the holes, during cutting of EPS. In Fig. 6 it is better shown that the temperature of the wire is increasing almost linearly, at the instants where the wire is inside the holes. The increase in temperature is local and is based on the absence of EPS material (to be melted) in the holes, where only the natural convection of air exerts a cooling effect on the wire. The temperature during cutting is determined by the first temperature-read-point inside the holes, where the temperature starts rising linearly, since at that instant the wire locally exits the EPS material by means of entering the cut holes. The dashed-line approximates the average cutting temperature for the particular cut experiment.

The method used for determining the kerfwidth, shown in Fig. 7, was first suggested by Brooks [9]. It relies on the fact that cut-paths are well determined by the robot positions and that cutting is performed with the same cutting parameters – producing equal kerfwidths. Though Brooks does not mention the method for measuring the cut-sample-height ( $h_s$ ), in the current work an electronic height-gauge (Mitutoyo Code. nr 192-651) was used.

Fig. 8 shows the relation between kerfwidth and temperature. The wire diameters are provided in swg units (Standard Wire Gauge) and correspond to  $25\text{swg} = D_{wire} = 0.508$  mm and  $20\text{swg} = D_{wire} = 0.914$  mm. Based on the fitting curve in Fig. 8, the kerfwidth was expressed as  $F_{kerf}(T) = 0.017615 T - 3.632$ , where the kerfwidth is measured in mm and temperature in degrees Celsius. We assume that the scatter in the experimental points is caused by randomly distributed macroscopic imperfections and small amount of water vapor in the EPS material. These imperfections are well known for industry and scientists, as f.ex. Grosvenor [20] already in the 60's refers to them as local wet and hard spots. According to [10] the EPS composition can cause a deviation in the melting energy of the foam. In the measurements of the melting energy a standard deviation of 10% was observed for a foam with the same composition. Moosa [19] suggests the density within a single piece of EPS material can change as much as 30–40% in some particular foams. Based on our pre-measurements and experience, the density of our foam does not vary significantly, however the previously mentioned defects and imperfections are certainly still present. For this reason, the current numerical code is used for general determination of the process parameters, as attempts are being made for increasing the accuracy and achievement of a more precise prediction of the kerfwidth.

## 5. Investigation of the angle of cutting

Another important part of the experiment is to test hot-wire



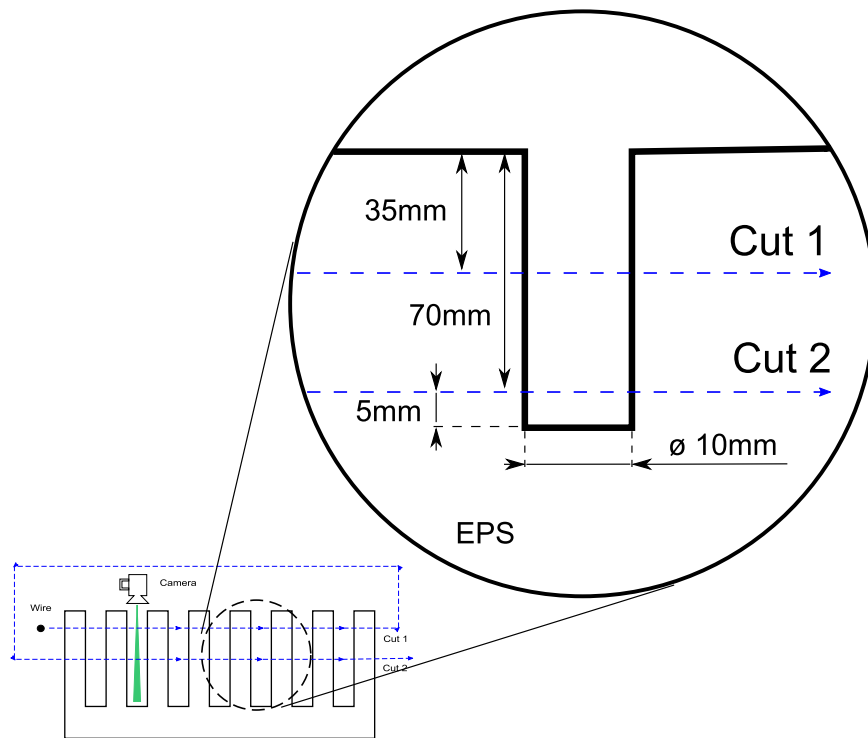


Fig. 4. Close view of the cut holes in EPS material used for visual observation of the wire during cutting.

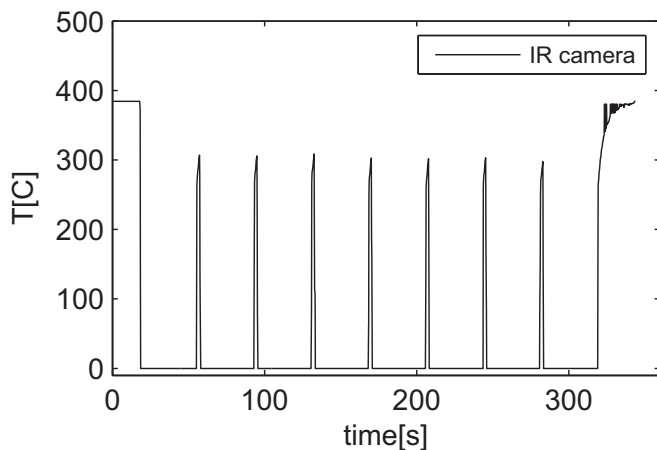


Fig. 5. Temperature of the hot-wire measured during cutting of EPS 150. Parameters:  $D_{wire} = 25\text{swg} = 0.508\text{ mm}$ ,  $I = 3.7\text{ A}$ , Speed = 5 mm/s. The temperature is measured only in the specially cut holes, shown in Fig. 3, corresponding to the peaks on the figure. The high temperatures in the beginning and end of cutting shows the wire temperature in air.

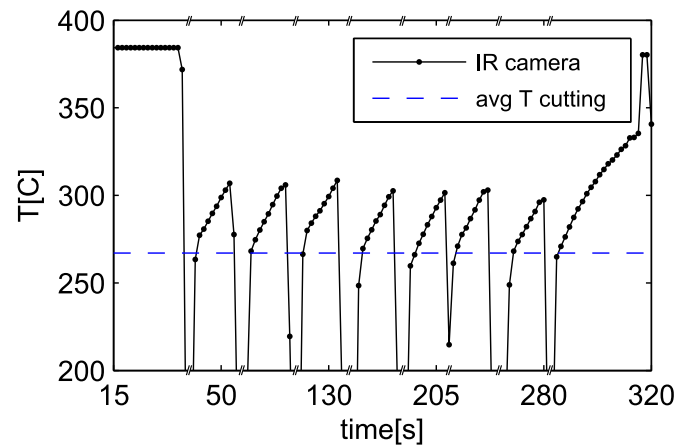
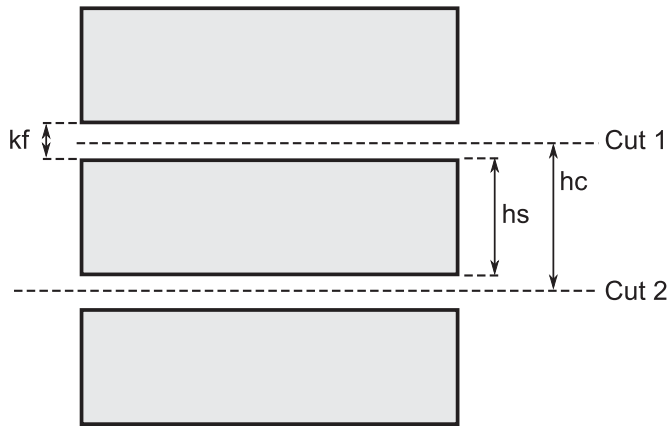


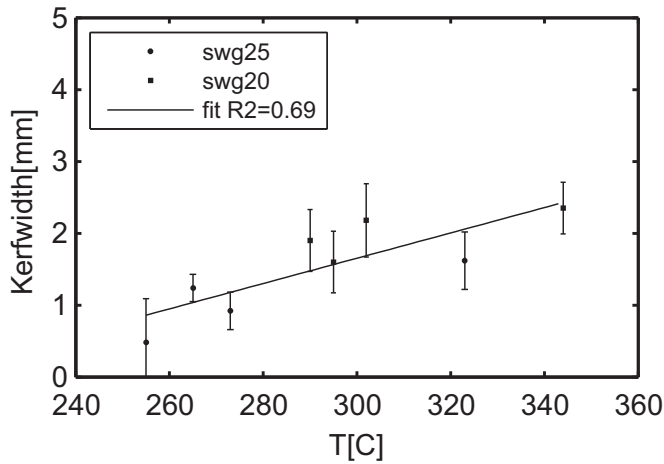
Fig. 6. Condensed data representation of Fig. 5 showing temperature of the hot-wire measured during cutting of EPS 150. Parameters:  $D_{wire} = 25\text{swg} = 0.508\text{ mm}$ ,  $I = 3.7\text{ A}$ , Speed = 5 mm/s. The data points are shown with dots. Parallel lines given at the x-axis indicate time-interruptions. The dashed line shows the approximated temperature of cutting being 267°C.

cutting of EPS with an angle toward the horizontal direction. The experiment aimed to investigate whether gravity would influence the kerfwidth. The results of the study are presented in Fig. 9, and it is shown that the kerfwidth is influenced by the angle of cutting with a particular trend valid for the both wire sizes used at the given constant cutting speed. The deviation from the horizontal cutting ( $0^\circ$ ) follows a well defined pattern and can be used for practical applications in industry or to draw analytical relations for further research. Though these results are very initial and valid only for these particular cutting scenarios they can be used as inspiration for future work where more accurate investigation of the parameters influencing the angle relations are done, as those can be introduced in the simulations algorithm by defining a new parameter accounting for the angle of cutting. Fig. 10 shows the

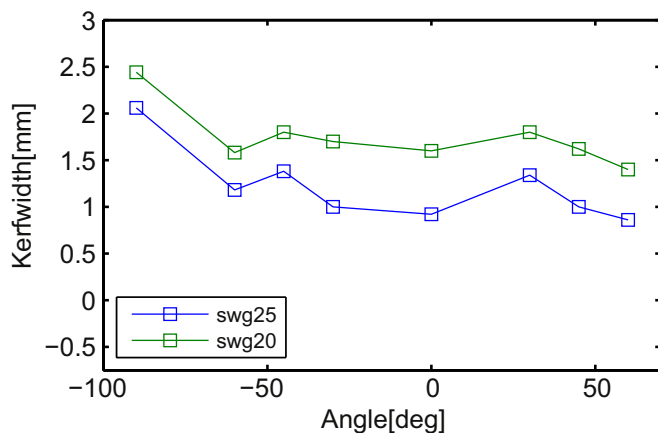
robotic set-up used during an experimental cutting of EPS at an angle of  $60^\circ$ . The hot-wire is not visible, due to its size and since it is inside the EPS, whereas the kerfwidth can be seen as a thin dark line on the side of the cut-sample close to the frame. The dark-line (i.e. the kerfwidth producing a cut into the sample) opens up when approaching toward the beginning of the sample probably due to small residual stresses in the material and for this reason the kerfwidth determination can be made only after the cutting procedure is completed. As shown in Fig. 10 the IR camera attached to the tool frame is always perpendicular to the surface of the EPS sample for visual observation of the hot-wire through the cut-holes. The cut-holes are shown in close, in Fig. 11, and the position and attachment of the IR camera to the tool frame is also visible. The samples for angle cutting are positioned on angle holders with



**Fig. 7.** Figure representing the measurement of kerfwidth based on the robot paths and samples measured after cutting. The figure shows a cross-section of an EPS block being cut with 2 hot wire cuts. The symbols  $k_f$ ,  $h_s$  and  $h_c$  denote the kerfwidth of single cutting, the height of the sample after cutting and the height of cutting path, where  $k_f = h_c - h_s$ .



**Fig. 8.** Kerfwidth with respect of temperature for different wire diameters.



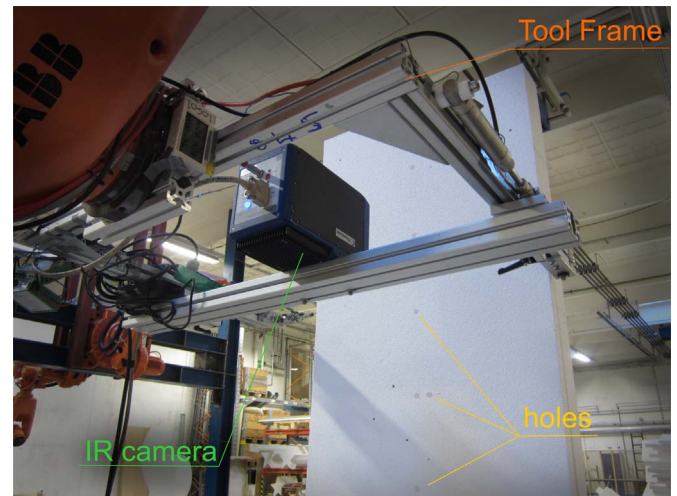
**Fig. 9.** Kerfwidth with respect to the cutting angle for different wire diameters. Negative angles denote cutting downward. The cutting was performed with parameters as: for wire 25swg- $I = 3.7A$ , Speed=5.0 mm/s, for wire 20swg- $I = 7.9A$ , Speed=5.0 mm/s. (For interpretation of color lines in this figure, the reader is referred to the web version of this article.)

different inclination allowing for cutting both up and down wards.

The increase of the wire diameter, due to the applied paint on the surface, was not more than 0.1 mm and for this reason it was



**Fig. 10.** A single experiment for robotic hot-wire cutting of an EPS sample with a 60° angle with respect to the horizontal plane. The IR camera is attached to the tool-frame.

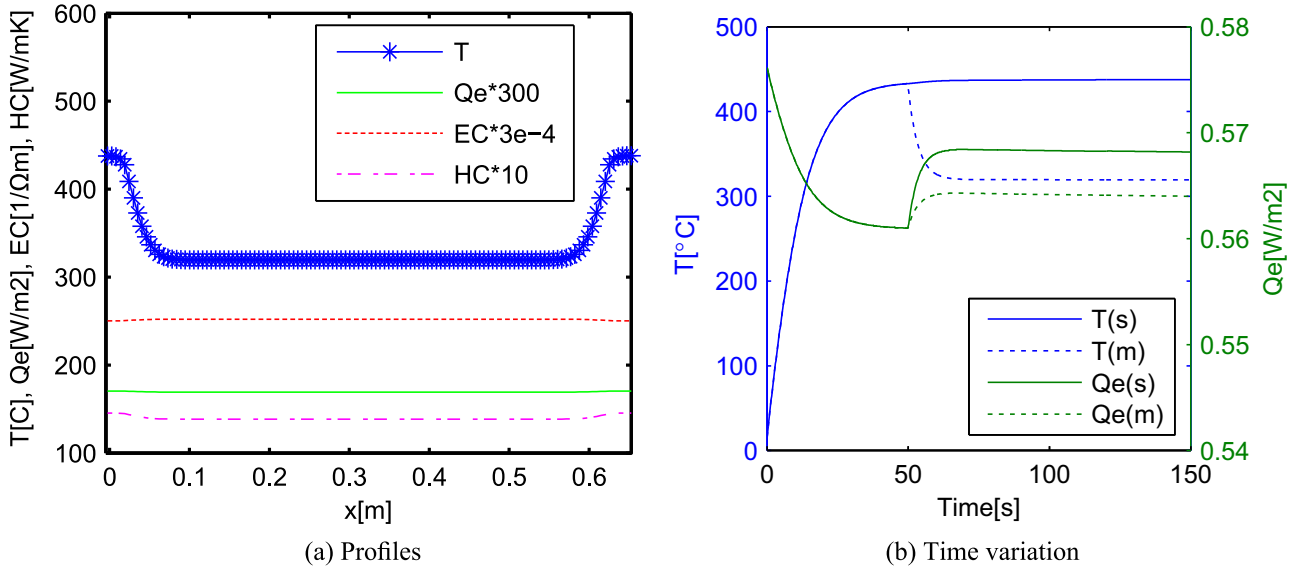


**Fig. 11.** A single experiment for robotic hot-wire cutting of EPS sample.

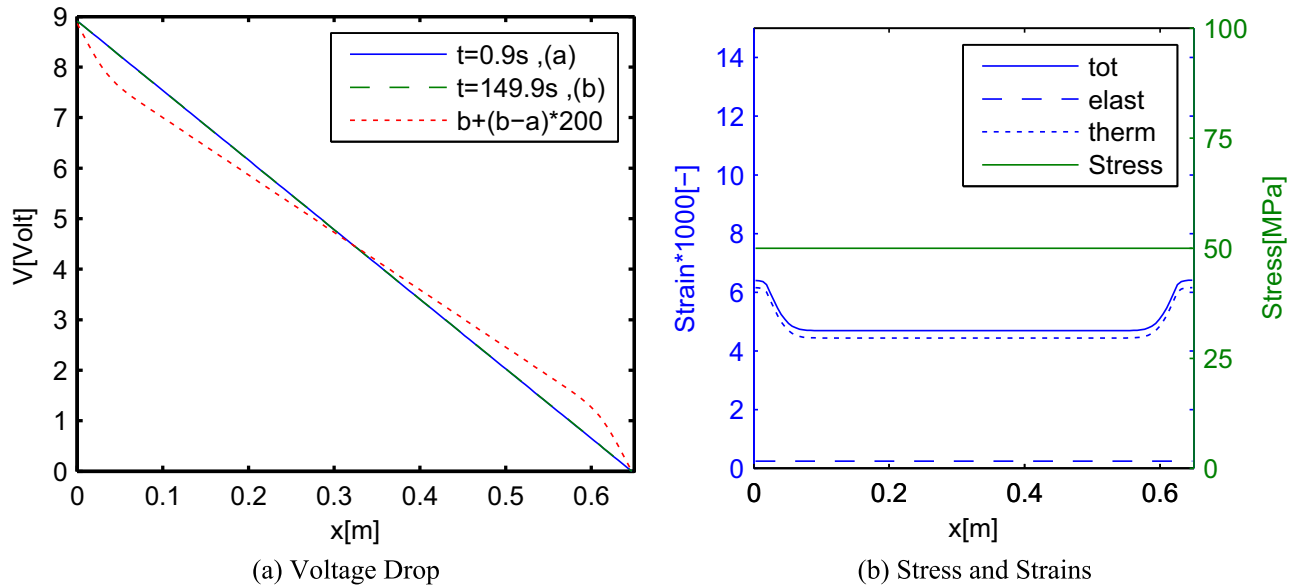
disregarded in this work.

## 6. Results

The output of the numerical model is expressed visually in Figs. 12–14. The values of the different parameters – both time and space dependent – are calculated and plotted. Figs. 12–14 represent a single simulation describing the process of a hot-wire cutting of EPS150 lasting for a time of 150 s. During the first 50 s



**Fig. 12.** The figures show temperature and various parameters with respect to space and time for the first 150 s of the cutting process. (a) shows temperature (T), generated energy through resistance heating (Qe), electrical conductivity (EC) and thermal conductivity (HC) with respect to position x (space) at  $t \approx 150$  s. (b) shows temperature (T) and Qe with respect to time for two points- "s" and "m" as shown in Fig. 1. Process parameters for wire 20swg –  $I = 7.4A$ , speed = 4 mm/s. (a) Profiles. (b) Time variation. (For interpretation of color lines in this figure, the reader is referred to the web version of this article.)



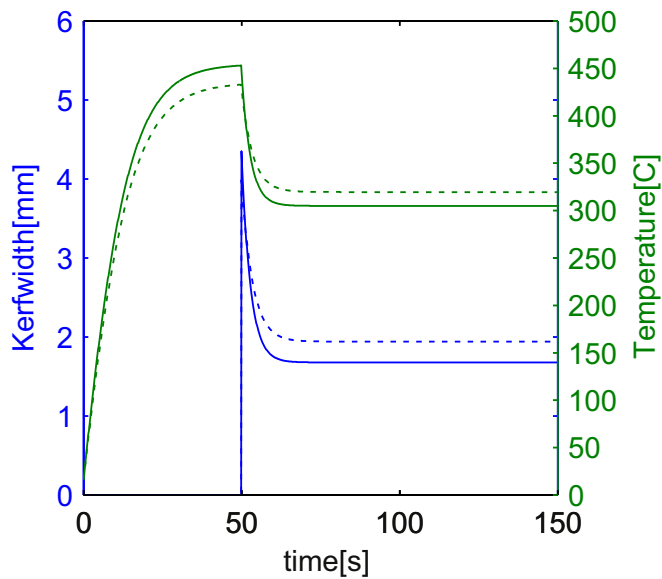
**Fig. 13.** The figure shows wire strain and wire voltage distribution with respect to space. (a) Shows the voltage drop through the wire with respect to x(space) at two time instants. The difference between the profiles at the time instants is given as a function of  $b + (b - a) * 200$ . (b) Shows total, elastic and thermal strains with respect to position x. Process parameters for wire 20swg –  $I = 7.4A$ , speed = 4 mm/s. (a) Voltage Drop. (b) Stress and Strains. (For interpretation of color lines in this figure, the reader is referred to the web version of this article.)

the wire is heated up in air environment and then it is brought in contact with the EPS (i.e. cutting the material) during the remaining 100 s. Fig. 12b shows the temperature and the generated energy ( $\dot{Q}_e$ ) due to electrical heating at two points on the wire – Point "s"(side) and Point "m" (middle), see Fig. 1. The two presented points are carefully chosen, so they describe the behavior of the wire in the two domains (or environments) in which the wire is operating during the cutting of EPS lasting 100 s. As we can see on the figure, the wire sides in air-(s) have a higher temperature. The reason for the higher temperature is the lower cooling effect of the air compared to the cooling produce by melting of the EPS. The higher temperature also increases the local electrical resistivity and for this reason the air-exposed parts of the wire generate more power through electrical heating ( $\dot{Q}_e$ ), seen Fig. 12a to

observe the local field variations.

Fig. 12a shows the variation of the calculated fields through-out the length of the wire during cutting. Based on the computed temperature variation along the wire, we can simulate the variation of the kerfwidth with respect to position of the wire in the EPS- an effect well known in practice, however never simulated previously. Thermal and electrical conductivity of the wire are also plotted with respect to space and can be used for process optimization.

Fig. 13 shows the space dependent stress and strains in the wire and the voltage drop. The wire is kept straight at all times with a pressure piston, such that the physical expansion of the wire as a function of applied power, temperature and cutting speed and the resulting stress relaxation is counteracted by dynamically keeping



**Fig. 14.** Figure showing the simulated temperature and kerfwidth with respect to time for simulations E1-solid line (20swg –  $I = 7.9A$ , speed = 5 mm/s), and E3-dashed line (20swg –  $I = 7.4A$ , speed = 4 mm/s). (For interpretation of color lines in this figure, the reader is referred to the web version of this article.)

**Table 1**

Table showing model parameters for both- Ni–Cr 20% resistance wire and expanded polystyrene foam (EPS). The value for  $H_f$  is experimentally determined as part of the present work.

| Parameters                            |             |                                       |
|---------------------------------------|-------------|---------------------------------------|
| Parameter                             | Magnitude   | Unit                                  |
| L                                     | 0.65        | [m]                                   |
| L2                                    | 0.60        | [m]                                   |
| $D_w$                                 | 0.508–0.914 | [mm]                                  |
| $\rho$                                | 8400        | [kg/m <sup>3</sup> ]                  |
| E                                     | 200         | [GPa]                                 |
| $\alpha$                              | 14.5        | [ $\mu\text{m}/\text{K}\mu\text{m}$ ] |
| $T_{\text{air}} = T_{\text{initial}}$ | 16.1        | [C]                                   |
| $H_f$                                 | 536,400     | [J/kg]                                |
| $\rho_f$                              | 25          | [kg/m <sup>3</sup> ]                  |
| $\epsilon$                            | 0.94        | [–]                                   |

the tensile stress in the wire at a constant level. Moreover the predicted wire strains provide information about the mechanical condition of the wire, its local expansion and total length, which is useful in cases when the thermal expansion has to be compensated for. The voltage distribution with respect to space is closely related to the generated energy ( $\dot{Q}_e$ ) and gives an in-dept understanding about the electrical-power relations in the wire. This feature is extremely useful in cases where the wire is subjected to space-dependent velocities (i.e. cutting with rotation).

**Table 2**

Comparison between experimental and simulated results for temperatures and kerfwidth during cutting of expanded polystyrene foam(EPS) with Ni–Cr 20% hot-wire. Speed and Temperature are measured in units m/s and °C, respectively. The results are representation of the conditions during a steady state cutting.

| No. | swg | (A) | (V)   | Speed  | Exp T   | Sim T | error % | Exp kf | Sim kf | error % |
|-----|-----|-----|-------|--------|---------|-------|---------|--------|--------|---------|
| S1  | 25  | 3.7 | 14.58 | 0.0050 | 265–276 | 283.0 | 4.6%    | 0.92   | 1.28   | 39%     |
| S2  | 25  | 3.7 | 14.58 | 0.0040 | 315–325 | 300.7 | 6.0%    | 1.62   | 1.60   | 1%      |
| S3  | 25  | 3.4 | 13.61 | 0.0040 | 260–271 | 289.1 | 8.9%    | 1.24   | 1.39   | 12%     |
| S4  | 25  | 3.7 | 14.58 | 0.0055 | 255–265 | 276.6 | 6.4%    | (0.48) | 1.17   | 144%    |
| S5  | 20  | 7.9 | 9.29  | 0.0050 | 280–305 | 305.0 | 4.3%    | 1.6    | 1.68   | 5%      |
| S6  | 20  | 7.9 | 9.29  | 0.0048 | 295–308 | 308.8 | 2.4%    | 1.90   | 1.75   | 8%      |
| S7  | 20  | 7.4 | 8.95  | 0.0040 | 290–315 | 319.6 | 5.7%    | 2.18   | 1.94   | 11%     |

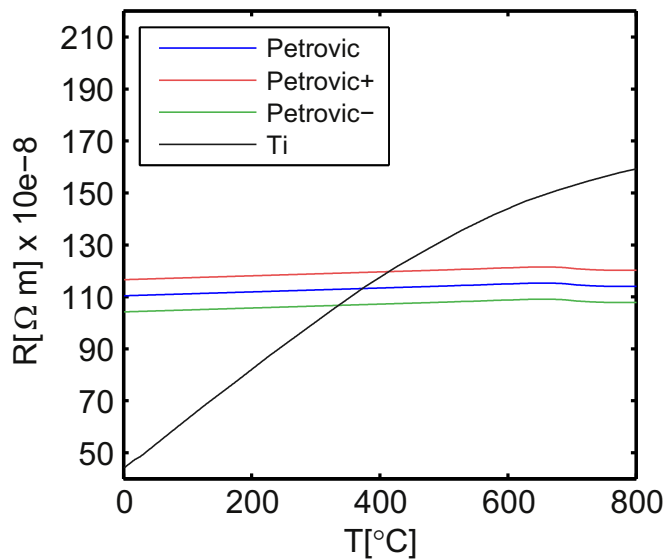
The predicted mid-wire kerfwidth is presented in Fig. 14 as a function of time. At the very initial stage of cutting, when the wire is brought into contact with the EPS ( $t = [50–52]$ s), the kerfwidth is high for a short moment due to the high temperature of the wire. However, with the melting of the EPS at this high temperature the wire is subsequently cooling fast in an exponential manner at time  $t = [52–60]$ s, until it reaches an equilibrium steady-state cutting condition.

A comparison between experimental and simulated results is given in Table 2. The table presents the measured and simulated results for temperatures and kerfwidths for a number of simulations denoted with index “SX”, where X is the number of the simulation. The experimental temperatures (Exp T) are presented in their complete range of values to show the fluctuations expected during the cutting of the samples. The relative errors are calculated with the formula  $\text{error} = 100 \cdot \text{abs}((\text{Exp} - \text{Sim})/\text{Exp})$ , where in the case of the ranges of the experimental temperatures the average value is used, i.e.  $\text{error} = 100 \cdot \text{abs}((\text{avr}(\text{Exp}) - \text{Sim})/\text{avr}(\text{Exp}))$ . Table 2 shows that the simulated temperatures are in a good agreement with the measured values, as the relative error is below 10% for the investigated experimental set. The relative error in the predicted kerfwidth is varying more, with some values very close to the experimentally determined ones and others further away. However most of the values show errors below 15%, with only two values in sim S1 and S4 being far from the measurements. In the case of S4, which is the measurement performed with the highest speed, it is suspected that a slight drag force was influencing the wire and this probably negatively influenced the kerfwidth. The large deviations in S1 and S4 can also be caused by the inaccuracies already observed in the experimentally measured kerfwidth-temperature relation presented in Fig. 8.

## 7. Parametric study

In this section the influence of process parameters on the predicted results is discussed. Voltage, speed and wire dimensions are parameters externally defined by the user and will not be discussed in this section. The parametric study will focus on the internal parameters – temperature dependent electrical conductivity and EPS melting energy. The temperature dependent electrical conductivity of the Ni–Cr wire is obtained from Petrovic [17], for a temperature range of [67;1050]C. However during casting of the Ni–Cr alloy, the weight percent of the individual chemical elements can slightly differ from the desired composition and ultimately influence the electrical properties of the wire. Based on measurements for the wire resistance at room temperature, it was found that the resistivity of the wires used in the experiment was slightly higher than the same predicted by Petrovic [17]. The opposite variation was previously observed in [15], where the resistivity was found to be lower with  $0.062 \mu\Omega\text{m}$ . These variations in the wire resistivity had an influence on the measured currents during experiments and later on the simulated results. To





**Fig. 15.** Electrical resistivity with respect to temperature for Ni-Cr20 wire. Petrovic's data [17] (Petrovic), Petrovic's data off-set with  $+0.062 \mu\Omega\text{m}$  (Petrovic+) and Petrovic's data off-set with  $-0.062 \mu\Omega\text{m}$  (Petrovic-). The electrical resistivity of Titanium (Ti) based on [21] is provided for comparison reasons. (For interpretation of color lines in this figure, the reader is referred to the web version of this article.)

investigate the influence of electrical resistivity variation, three suggested curves for the temperature dependent electrical resistivity based on the measurements of Petrovic, were used – the original measured by Petrovic [17] and two curves of Petrovic's off-set with  $+0.062 \mu\Omega\text{m}$  ( $P+$ ) and  $-0.062 \mu\Omega\text{m}$  ( $P-$ ). Fig. 15 shows the temperature dependent resistivities of the three suggested curves for the temperature dependent electrical resistivity.

The melting energy of the EPS is another parameter that needs special consideration. The EPS material, used during the cutting

process, is melting and partially evaporating and for this reason the exact energy dissipated from the wire to the EPS is not well known. The study of Aitchison [10], discussed this topic, and it was found that the volumetric melting energy of the EPS during cutting was  $7.71 \text{ MJ/m}^3$ , with a standard deviation of 10%. This finding implies that the melting energy per mass is  $308400 \text{ J/kg}$  for EPS of density  $25 \text{ kg/m}^3$ . However during our study, it was found that the melting energy per mass is approximately  $536400 \text{ J/kg}$ , or 43% higher than the values provided by Aitchison. The difference between the measurements can be based on potential difference in the material composition of EPS (provided by different material suppliers), a slightly increased amount of water vapor in the material, or a different degree of evaporation of the EPS material. To give a quantitative measure of the importance of these parameters, simulations with decreased and increased melting energy of 4% combined with a number of expected curves for the temperature dependent electrical resistivity were simulated and compared with the experimental results in Tables 3 and 4. From the simulated results in these tables, it is clear that the cutting temperature and the kerfwidth increase with decrease of electrical conductivity and with decrease in melting energy. Moreover small deviations in both parameters (el. resistivity and melting energy) have a considerable influence on the results, as the max difference between "Sim P-/E-" (decreased conductivity and melting energy increased) and "Sim P+/E+" (increased conductivity and melting energy increased) is approx.  $21.4^\circ\text{C}$  for temperature and  $0.39 \text{ mm}$  for the kerfwidth.

## 8. Conclusions

A new numerical model for hot-wire cutting of expanded polystyrene is proposed. The model shows to be very useful for understanding the processes occurring in the cutting tool, i.e.: the heated wire. A number of process parameters such as –

**Table 3**

Comparison between experimental and simulated results for kerfwidth produced during cutting of expanded polystyrene foam (EPS) with Ni-Cr 20% hot-wire. A  $3 \times 3$  parameters variation of changing conductivity and melting energy ( $-4\%$ ,  $0\%$ ,  $+4\%$ ) resulting in 9 simulations was performed. Speed and Temperature are measured in units m/s and  $^\circ\text{C}$ , respectively. Simulations abbreviated with "E-"/"E+" are made with decreased/increased melting energy of 4%. Abbreviations "P", "P+", "P-" denote electrical resistivity according to Petrovic [17], Petrovic off-set with  $+0.062 \mu\Omega\text{m}$  and Petrovic off-set with  $-0.062 \mu\Omega\text{m}$ , respectively. The results are representation of the conditions during a steady state cutting.

| No. | Exp kf | Sim P+/E | Sim P+/E- | Sim P/E | Sim P/E- | Sim P-/E | Sim P-/E- | Sim P+/E+ | Sim P/E+ | Sim P-/E+ |
|-----|--------|----------|-----------|---------|----------|----------|-----------|-----------|----------|-----------|
| S1  | 0.92   | 1.28     | 1.33      | 1.35    | 1.41     | 1.43     | 1.49      | 1.23      | 1.30     | 1.37      |
| S2  | 1.62   | 1.60     | 1.66      | 1.69    | 1.75     | 1.78     | 1.86      | 1.54      | 1.62     | 1.71      |
| S3  | 1.24   | 1.39     | 1.45      | 1.47    | 1.53     | 1.55     | 1.61      | 1.34      | 1.41     | 1.49      |
| S4  | (0.48) | 1.165    | 1.21      | 1.23    | 1.28     | 1.30     | 1.35      | 1.12      | 1.18     | 1.25      |
| S5  | 1.6    | 1.68     | 1.75      | 1.77    | 1.84     | 1.87     | 1.95      | 1.61      | 1.70     | 1.80      |
| S6  | 1.90   | 1.75     | 1.80      | 1.84    | 1.92     | 1.95     | 2.03      | 1.68      | 1.77     | 1.87      |
| S7  | 2.18   | 1.94     | 2.01      | 2.05    | 2.13     | 2.17     | 2.25      | 1.86      | 1.97     | 2.08      |

**Table 4**

Comparison between experimental and simulated results for wire temperature during cutting of expanded polystyrene foam (EPS) with Ni-Cr 20% hot-wire. A  $3 \times 3$  parameters variation of changing conductivity and melting energy ( $-4\%$ ,  $0\%$ ,  $+4\%$ ) resulting in 9 simulations was performed. Speed and Temperature are measured in units m/s and  $^\circ\text{C}$ , respectively. Simulations abbreviated with "E-"/"E+" are made with decreased/increased melting energy of 4%. Abbreviations "P", "P+", "P-" denote electrical resistivity according to Petrovic [17], Petrovic off-set with  $+0.062 \mu\Omega\text{m}$  and Petrovic off-set with  $-0.062 \mu\Omega\text{m}$ , respectively. The results are representation of the conditions during a steady state cutting.

| No. | Exp kf  | Sim P+/E | Sim P+/E- | Sim P/E | Sim P/E- | Sim P-/E | Sim P-/E- | Sim P+/E+ | Sim P/E+ | Sim P-/E+ |
|-----|---------|----------|-----------|---------|----------|----------|-----------|-----------|----------|-----------|
| S1  | 265–276 | 283.0    | 285.8     | 286.9   | 289.8    | 291.2    | 294.3     | 280.2     | 283.9    | 288.0     |
| S2  | 315–325 | 300.7    | 304.2     | 305.5   | 309.2    | 310.8    | 314.8     | 297.1     | 301.7    | 306.9     |
| S3  | 260–271 | 289.1    | 292.2     | 293.3   | 296.6    | 298.0    | 301.4     | 286.1     | 290.1    | 294.6     |
| S4  | 255–265 | 276.6    | 279.1     | 280.1   | 282.8    | 284.0    | 286.9     | 274.0     | 277.4    | 281.1     |
| S5  | 280–305 | 305.0    | 308.7     | 310.0   | 313.9    | 315.7    | 319.8     | 301.3     | 306.1    | 311.5     |
| S6  | 295–308 | 308.8    | 312.7     | 314.1   | 318.1    | 319.9    | 324.2     | 305.0     | 310.0    | 315.6     |
| S7  | 290–315 | 319.6    | 323.8     | 325.4   | 329.9    | 331.9    | 336.7     | 315.3     | 320.9    | 327.1     |

temperature, generated energy for electrical heating, voltage distribution in the wire, stress, strains and kerfwidth are predicted based on modeling and experimental findings. Consideration regarding cutting with different horizontal angles is proposed and discussed. The accuracy of the model is determined by the experimental findings relating kerfwidth and temperature. The sources of uncertainty based on the variations in the electrical resistivity and the EPS melting energy are discussed in short, as these are to be further investigated in future works.

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